PROBLEM 1.2

Malthusian Population Dynamics & Doubling Time

Original statement · ProblemsCh1.pdf

Separable ODE • Asymptotics • Doubling Time

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: Let $x(t)$ be the population of a certain species as a function of time, and suppose that the rate of change of this population is proportional to the existing population at each time, with constant of proportionality $k \in \mathbb{R}$.

(i) Write a differential equation for $x(t)$.
(ii) Solve it, determine the behavior of $x(t)$ as $t \to \infty$ and interpret it depending on the sign of k.
(iii) If the initial population is $x(0) = x_0$ and $k > 0$, determine the time needed for the population to double.

PHASE 2: FUNDAMENTAL FORMULATION

The rate of change is the time derivative $\frac{dx}{dt}$. Proportionality to $x(t)$ with constant k yields:

$$\frac{dx}{dt} = kx(t), \quad x(0) = x_0 > 0$$

This is a first-order linear separable homogeneous ODE: the population equation of the class notes.

METHOD OF THE CLASS NOTES [NOTES, P. 2]

Since $x(t) > 0$, divide by x and integrate both sides from 0 to t; evaluate the left side either with the substitution $u = x(\tau)$ or by recognising $\big(\ln x(\tau)\big)'$.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): By definition, the ODE is:

$$\boxed{\frac{\mathbf{dx}}{\mathbf{dt}} = \mathbf{kx}} \qquad (1.2.1)$$

Part (ii): The population is positive, $x(t) > 0$, so divide by it, $\frac{x'(\tau)}{x(\tau)} = k$, and integrate both sides from 0 to t [Notes, p. 2]:

$$\int_0^t \frac{x'(\tau)}{x(\tau)}\, d\tau = \int_0^t k\, d\tau$$

Left side. Substitute $u = x(\tau)$, $du = x'(\tau)\, d\tau$; the limits $\tau = 0, \tau = t$ become $u = x_0, u = x(t)$. An antiderivative of $1/u$ ($u > 0$) is $\ln u$, so by Barrow's rule

$$\int_{x_0}^{x(t)} \frac{du}{u} = \Big[\ln u\Big]_{x_0}^{x(t)} = \ln x(t) - \ln x_0 = \ln\left(\frac{x(t)}{x_0}\right).$$

(Second route of the notes: by the chain rule $\frac{d}{d\tau} \ln x(\tau) = \frac{x'(\tau)}{x(\tau)}$, so the integrand is $\big(\ln x(\tau)\big)'$.) *Right side.* $\int_0^t k\, d\tau = [k\tau]_0^t = kt$. So $\ln\left(\frac{x(t)}{x_0}\right) = kt$. Exponentiate both sides:

$$\boxed{\frac{x(t)}{x_0} = e^{kt} \implies \mathbf{x(t) = x_0 e^{kt}}} \qquad (1.2.2)$$

Asymptotic limits as $t \to \infty$ (the notes: "if $k > 0$ population increases, if $k < 0$ it decreases"; the case $k = 0$ is added here):

- If $k > 0$: $\lim_{t\to\infty} x(t) = \lim_{t\to\infty} x_0 e^{kt} = +\infty$ (Exponential Malthusian explosion; natality exceeds mortality).
- If $k = 0$: $x(t) = x_0 e^0 = x_0$ (Static equilibrium; births perfectly balance deaths).
- If $k < 0$: $\lim_{t\to\infty} x(t) = \lim_{t\to\infty} x_0 e^{-|k|t} = 0$ (Exponential extinction; mortality exceeds natality).

Part (iii): Doubling Time t_d:

Set $x(t_d) = 2x_0$ with $k > 0$:

$$x_0 e^{kt_d} = 2x_0 \implies e^{kt_d} = 2$$

Take the natural logarithm:

$$kt_d = \ln 2 \implies \mathbf{t_d = \frac{\ln 2}{k}} \tag{1.2.3}$$

PHASE 4: FINAL RESULT & DIMENSIONAL VERIFICATION

BOXED RESULT

$$x(t) = x_0 e^{kt}, \quad t_d = \frac{\ln 2}{k}$$

Dimensional Verification: $[k] = \text{time}^{-1} \implies [t_d] = \frac{1}{[k]} = \text{time}$, which is dimensionally homogeneous.